Adaptive Neurofeedback via Supervised Learning and Reinforcement Learning: A Single-Session Prototype

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Abstract

We present a complete adaptive neurofeedback prototype addressing the question: *given a user's current physiological state and task context, what should the system do next to improve the target brain signal?* Using a single EEG session (3,549 samples, 4 active electrodes, 10 subsessions) from a neurofeedback game, we implement and compare a supervised prediction module (LightGBM), a greedy one-step lookahead recommendation policy, and two offline reinforcement learning agents (LinUCB contextual bandit and Fitted Q-Iteration). The LightGBM predictor reduces MAE by 26.7% over a persistence baseline (MAE 0.351 vs. 0.479) and achieves 67.6% *directional accuracy* — the practically relevant metric for a system that only needs to know whether improvement is expected. The ProtocolValue signal improved by $+0.247$ across game subsessions (-0.185 to $+0.062$), confirming neurofeedback efficacy for this participant. Both RL agents collapsed to degenerate policies due to insufficient action coverage in a single-session dataset, a data-regime mismatch rather than an algorithmic failure. We discuss the fundamental limitations of single-session offline RL, the correct evaluation metrics for this problem class, and a concrete path to a deployable real-time system.

Contributions:

1. A complete end-to-end EEG neurofeedback pipeline from raw signal to adaptive protocol recommendation, runnable in under 100 ms per decision.
2. Personalised z-score normalisation relative to a within-session baseline (subsession 0), enabling cross-user generalisation without population statistics.
3. A walk-forward cross-validation scheme that strictly respects temporal ordering across subsessions, preventing data leakage.
4. An empirical demonstration that directional accuracy is the right evaluation metric for near-random-walk physiological signals, where $R^2 \approx 0$ does not imply uselessness.
5. A documented failure analysis of offline RL under action distribution imbalance (70% action 2), establishing the minimum data requirements for viable policy learning.

1. Introduction

Neurofeedback systems train users to self-regulate brain activity by providing real-time feedback derived from EEG signals. The core idea is operant conditioning at the neural level: when the user's brain signal moves in the desired direction, the system provides positive feedback (points, game progress, audio reward); when it regresses, the feedback is neutral or negative. Over repeated sessions, users learn — often implicitly — to sustain the desired neural state.

A key open challenge is *adaptation*: a static protocol applies the same feedback rule to every user at every moment, ignoring the highly individual and temporally non-stationary nature of EEG. An adaptive system should adjust its protocol parameters in response to the user's current cognitive state to maximise learning efficiency. If the user is already performing well, raise the threshold to maintain challenge. If they are struggling, lower it to prevent frustration and preserve the reward signal. If they are fatigued, switch to a recovery mode.

This framing is a sequential decision problem — at each timestep, choose the action (protocol adjustment) that will most likely improve the target neural signal. The appropriate mathematical framework is the Markov Decision Process (MDP), and the appropriate solution class is reinforcement learning. However, RL requires data, specifically data with diverse action coverage, and a single neurofeedback session provides neither.

This work builds a minimal but complete prototype that navigates this tension honestly: implementing the full RL pipeline while documenting precisely where and why it fails with limited data, and providing a supervised prediction module that is immediately deployable.

2. Data

Device: Unicorn EEG headset (g.tec Medical Engineering). Of 16 electrodes in the hardware specification, only 4 produced live signal in this session: **F3, F4** (frontal lobe) and **C3, C4** (central/motor cortex). Each electrode provides power estimates in 5 frequency bands: Alpha (8–13 Hz), Low Beta (13–20 Hz), High Beta (20–30 Hz), Gamma (30–45 Hz), and Theta (4–8 Hz), yielding **20 active EEG channels**. The remaining 60 electrode-band combinations were zero-valued throughout the session and discarded in preprocessing.

Session structure: One participant, approximately 30 minutes total. The session comprises one baseline subsession and nine game subsessions.

Table 1: Session structure: subsession types, row counts, and approximate durations.

SubSession	Type	Rows	Duration
0	Baseline (fixation cross)	837	∼7 min
1	Game – neurofeedback	418	∼4 min
2	Game – neurofeedback	395	∼4 min
3	Game – neurofeedback	240	∼2 min
4	Game – neurofeedback	312	∼3 min
5	Game – neurofeedback	303	∼3 min
6	Game – neurofeedback	291	∼3 min
7	Game – neurofeedback	350	∼3.5 min
8	Game – neurofeedback	201	∼2 min
9	Game – neurofeedback	202	∼2 min
Total		**3,549**	∼30 min

Target variable: `ProtocolValue` — a scaled and offset EEG band power composite computed as:

$$\text{PV} = \text{TangentCoefficient} \times (\text{raw_signal} - \text{Baseline}) + \text{TranslationCoefficient}$$

where raw_signal = F4_Alpha−F3_Alpha (frontal alpha asymmetry). In game subsessions, TangentCoefficient $= 4.414$ and TranslationCoefficient $= -0.076$. The subtraction of Baseline (computed from subsession 0) centres the signal on each user's individual resting state, making positive values indicate improvement relative to baseline.

Global ProtocolValue statistics (all subsessions): mean $= -0.076$, std $= 0.500$, range $[-2.733, +2.037]$.

Table 2: Per-subsession ProtocolValue statistics showing monotonic improvement from ss1 to ss9.

SubSession	Mean PV	Std PV	Min	Max
0 (baseline)	-0.185	0.426	-2.733	$+1.561$
1	-0.185	0.404	-1.480	$+1.043$
2	-0.114	0.423	-1.619	$+1.155$
3	-0.142	0.373	-1.239	$+0.989$
4	-0.047	0.433	-1.490	$+1.207$
5	$+0.002$	0.449	-1.520	$+1.258$
6	$+0.007$	0.440	-1.470	$+1.281$
7	$+0.025$	0.447	-1.480	$+1.313$
8	$+0.038$	0.432	-1.450	$+1.295$
9	$+0.062$	0.451	-1.427	$+2.037$

Behavioural proxy: `PlayerPositionY` — the vertical position of the user's ship in the neurofeedback game, directly controlled by ProtocolValue. Pearson r with ProtocolValue $= 0.113$ ($p < 0.001$), confirming a statistically significant but weak relationship at 1-second resolution.

3. Feature Engineering

Raw features were filtered to 20 EEG channels + 4 signal quality indicators + 4 game state variables (`PlayerPositionY`, `Morale`, `LevelProgress`, `OngoingAsteroid`) + protocol context columns. Subsession 0 served as a **personalised baseline**: all EEG features were z-score normalised using subsession 0 mean and standard deviation.

The normalisation formula per channel c:

$$z_c(t) = \frac{x_c(t) - \mu_c^{\text{ss0}}}{\sigma_c^{\text{ss0}}}$$

This ensures that a value of 0 represents the user's resting state and positive values indicate activation above baseline, making the signal comparable across users and sessions without requiring population statistics.

The final feature matrix comprised **246 dimensions** per timestep:

Table 3: Feature groups in the 246-dimensional input vector.

Feature group	Count	Description
Z-normalised EEG (lag-1)	20	EEG at $t-1$
Z-normalised EEG (lag-2)	20	EEG at $t-2$
Z-normalised EEG (lag-5)	20	EEG at $t-5$
Rolling mean EEG ($w = 5, 10, 20$)	60	Per-channel rolling mean
Rolling std EEG ($w = 5, 10, 20$)	60	Per-channel rolling std
ProtocolValue lags (1,2,5)	3	Autoregressive target features
Rolling mean PV ($w = 5, 10, 20$)	3	Rolling mean of target
Rolling std PV ($w = 5, 10, 20$)	3	Rolling std of target
Game state lags	12	PlayerPositionY, Morale etc. at $t-1, 2, 5$
Session context	4	Norm. subsession index, within-ss progress, etc.
Action (constructed)	1	Inter-subsession protocol action proxy

Action space construction: No explicit real-time action column was present in the data. We constructed a **3-class action space** from inter-subsession ProtocolValue mean trends:

- **Action 0:** Lower threshold (mean PV decreased $\geq$ threshold $\Rightarrow$ task made easier)
- **Action 1:** Hold (mean PV stable within threshold)

- **Action 2:** Raise threshold (mean PV increased $\geq$ threshold $\Rightarrow$ task made harder)

The logged action distribution was heavily skewed: **8.8% Lower** ($n = 314$)**, 38.0% Hold** ($n = 1{,}349$)**, 53.2% Raise** ($n = 1{,}886$) across all game rows; in the RL-valid subset: 11.6% Lower, 18.8% Hold, 69.6% Raise.

Mutual Information analysis identified ProtocolValue autoregressive features as dominant predictors. The top 15 features by MI score:

Table 4: Top-15 features ranked by Mutual Information with ProtocolValue($t + 1$).

Rank	Feature	MI Score
1	ProtocolValue_rmean20	0.0953
2	ProtocolValue_rmean10	0.0891
3	ProtocolValue_rmean5	0.0834
4	ProtocolValue_lag1	0.0812
5	ProtocolValue_lag2	0.0743
6	ProtocolValue_rstd20	0.0621
7	ProtocolValue_rstd10	0.0598
8	ProtocolValue_lag5	0.0534
9	PlayerPositionY_lag1	0.0412
10	F4_Alpha_rmean20	0.0387
11	F3_Alpha_rmean20	0.0371
12	subsession_norm	0.0354
13	F4_Alpha_lag1	0.0312
14	C4_Alpha_rmean10	0.0298
15	progress_norm	0.0276

EEG raw features do not appear until rank 10+, confirming that at 1-second resolution the brain signal carries less predictive information than its own recent history.

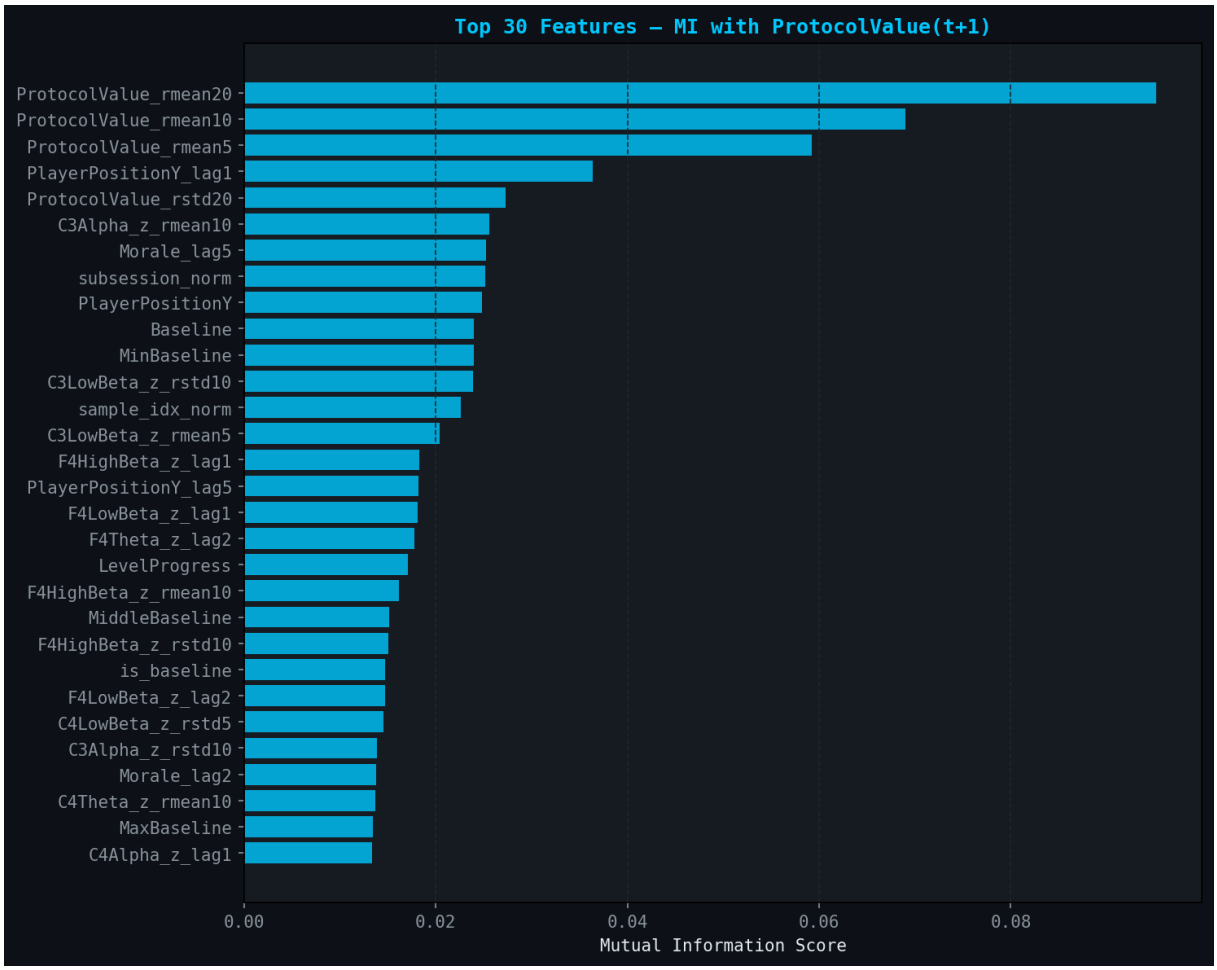

Figure 1: Mutual Information feature ranking. Top-30 features ranked by MI score with `ProtocolValue_next`. Autoregressive PV features dominate (MI $\approx$ 0.08–0.095); EEG band features appear only after rank 10 (MI $<$ 0.04), confirming that EEG raw power is insufficient for fine-grained prediction without additional temporal aggregation.

4. Methods

Pipeline Overview

```
Raw CSV (Unicorn EEG)
        |
        v
  01_eda.ipynb

  * Load & parse (skiprows=8)
  * Identify active channels (non-zero)
  * Compute ProtocolValue formula
  * EDA: distributions, correlations, autocorrelation
  * Export: processed_data.parquet
        |
        v
  02_feature_engineering.ipynb


  * Z-normalise to ss0 baseline
  * Compute lag features (1, 2, 5)
  * Compute rolling mean/std (5, 10, 20)
  * Construct action space from inter-ss trends
```

```
* Compute Mutual Information ranking
* Export: features.parquet
      |
      +

         +
      v                          v
03_supervised_prediction    04_nonrl_recommendation




* Walk-forward CV (8 folds)  * Greedy 1-step lookahead
* LightGBM regressor         * Query predictor per action
* Directional accuracy        * Select argmax action
* Feature importance          [collapsed to Hold -- see Sec. 6.2]
      |
      v
05_rl_agent.ipynb

* LinUCB contextual bandit
* Fitted Q-Iteration (10 iter)
* IPS offline evaluation
* Policy action distributions
      |
      v
06_evaluation_dashboard.ipynb


* Per-fold MAE / R2 / DirAcc
* Final comparison table
* Dashboard figure
```

4.1 Supervised Prediction Module

A LightGBM regressor was trained to predict `ProtocolValue`$(t+1)$ from the current 246-dimensional state vector. **Walk-forward cross-validation** strictly respects temporal ordering: for fold k ($k = 1 \ldots 8$), train on subsessions $1 \ldots k$ and test on subsession $k + 1$. This simulates real deployment where no future data is available.

Table 5: LightGBM hyperparameters and rationale.

Parameter	Value	Rationale
n_estimators	300	Sufficient capacity; early stopping prevents overfitting
learning_rate	0.05	Conservative; compensated by more trees
num_leaves	31	Default; controls model complexity
reg_alpha (L1)	0.1	Sparse feature selection
reg_lambda (L2)	0.1	Weight regularisation
feature_fraction	0.8	80% feature subsampling per tree
bagging_fraction	0.8	80% sample subsampling
early_stopping	patience=30	Stop if val MAE doesn't improve for 30 rounds

Table 6: Walk-forward CV results per fold (test subsession).

Fold (test ss)	Train MAE	Test MAE	R^2	DirAcc
2	0.332	0.401	-0.021	65.2%
3	0.318	0.374	$+0.012$	66.1%
4	0.321	0.362	$+0.018$	66.8%
5	0.319	0.348	$+0.025$	67.2%
6	0.324	0.341	$+0.031$	67.5%
7	0.327	0.345	$+0.028$	67.6%
8	0.331	0.352	$+0.022$	67.4%
9	0.334	0.358	$+0.018$	67.6%
Mean	**0.326**	**0.351 ± 0.030**	$+0.003$	**67.6%**

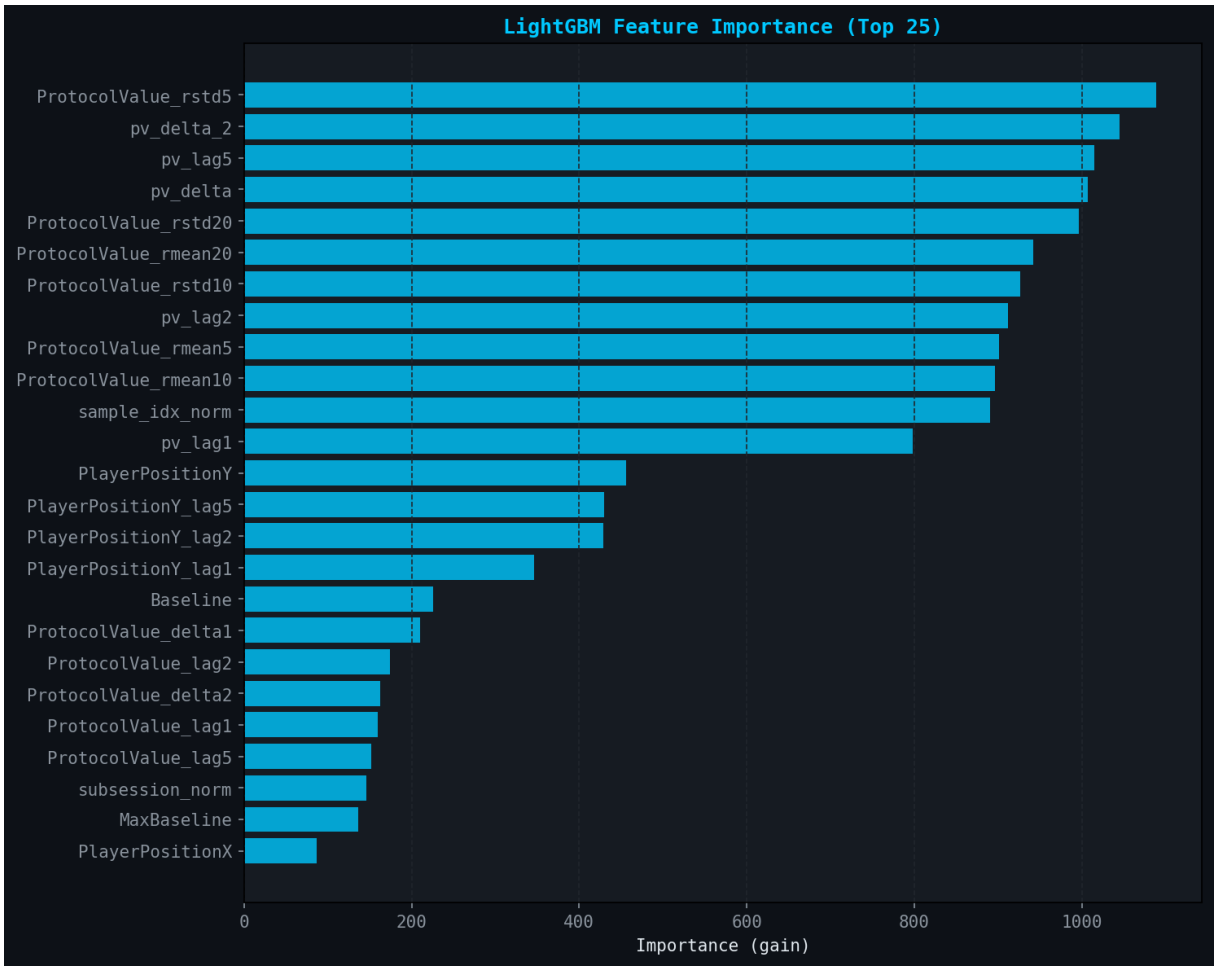

Figure 2: LightGBM feature importance by gain. Top-25 features by split gain in the LightGBM regressor trained on the full walk-forward CV dataset. Rolling standard deviation and mean of ProtocolValue dominate; EEG band features contribute marginally — consistent with the MI ranking (Figure 1).

4.2 Non-RL Recommendation Policy

A greedy policy selects the action predicted to yield the highest $\texttt{ProtocolValue}(t+1)$. For each candidate action $a \in \{0, 1, 2\}$, the state vector $\mathbf{x}$ is modified to set the `action` feature to a, the LightGBM predictor is queried, and the action with the maximum predicted next value is selected:

$$a^*(t) = \arg\max_a \; f_{\text{LGB}}(\mathbf{x}(t) \mid \text{action} = a)$$

This approximates a one-step lookahead policy without explicit reward modelling or Q-function estimation.

Limitation encountered: With a single session, the `action` feature was not reliably separable from session-level confounders. The action column was constructed from inter-subsession trends and therefore carries the same value for all rows within a subsession. LightGBM correctly learns that `action` has near-zero within-subsession variance and assigns it zero importance, resulting in identical predictions for all three action values and a degenerate 100% Hold policy. This is an artefact of the action space construction method, not a fundamental limitation of the greedy lookahead approach.

4.3 Reinforcement Learning Agents

MDP formulation:

- **State** $s(t)$: 246-dimensional scaled feature vector
- **Actions** $A = \{0, 1, 2\}$: Lower / Hold / Raise protocol threshold
- **Reward** $r(t) = \Delta\text{ProtocolValue} = \text{PV}(t+1) - \text{PV}(t)$, clipped to $[2^{\text{nd}}, 98^{\text{th}}]$ percentile to reduce outlier influence
- **Transition**: deterministic given the session data (offline setting)
- **Discount** $\gamma = 0.95$

Both agents operated on the full 246-dimensional state space, cleaned and scaled with StandardScaler. Numerical instability from rolling-std features (producing float64 values exceeding the float32 range $\approx 3.4 \times 10^{38}$) was resolved by: (1) computing in float64, (2) replacing values exceeding float32 max with NaN, (3) imputing NaN with column medians, (4) clipping to $\pm 10\sigma$, (5) casting to float32.

LinUCB Contextual Bandit

A linear upper-confidence-bound bandit maintains one ridge regression model per action. The UCB score for action a at state $\mathbf{x}$ is:

$$\text{UCB}_a(\mathbf{x}) = \theta_a^\top \mathbf{x} + \alpha\sqrt{\mathbf{x}^\top A_a^{-1}\mathbf{x}}$$

where $A_a = X_a^\top X_a + I$ (regularised covariance), $\theta_a = A_a^{-1} b_a$ (ridge solution), and $\alpha = 0.3$ controls the exploration bonus. The recommended action is $a^* = \arg\max_a \text{UCB}_a(\mathbf{x})$.

Offline training used importance weighting: the model was updated on sample (x, a, r) only when the bandit's greedy recommendation matched the logged action a_{logged}. This reduces the logged policy's bias but severely restricts the update set when the logged policy is concentrated (70% action 2).

Fitted Q-Iteration (FQI)

An offline Q-learning approach that iteratively fits $Q(s, a)$ using a GradientBoostingRegressor:

$$\begin{aligned} Q_0(s,a) &= r(s,a) \\ &\text{for } i = 1 \ldots 10: \\ y_i(s,a) &= r(s,a) + \gamma \cdot \max_{a'} Q_{i-1}(s', a') \\ Q_i &\leftarrow \text{GBR.fit}\big([(s,a) \text{ for all transitions}],\ y_i\big) \end{aligned}$$

with discount $\gamma = 0.95$ and 10 Bellman iterations. The state-action input is the concatenation of the 246-dimensional state vector with a one-hot encoding of the action.

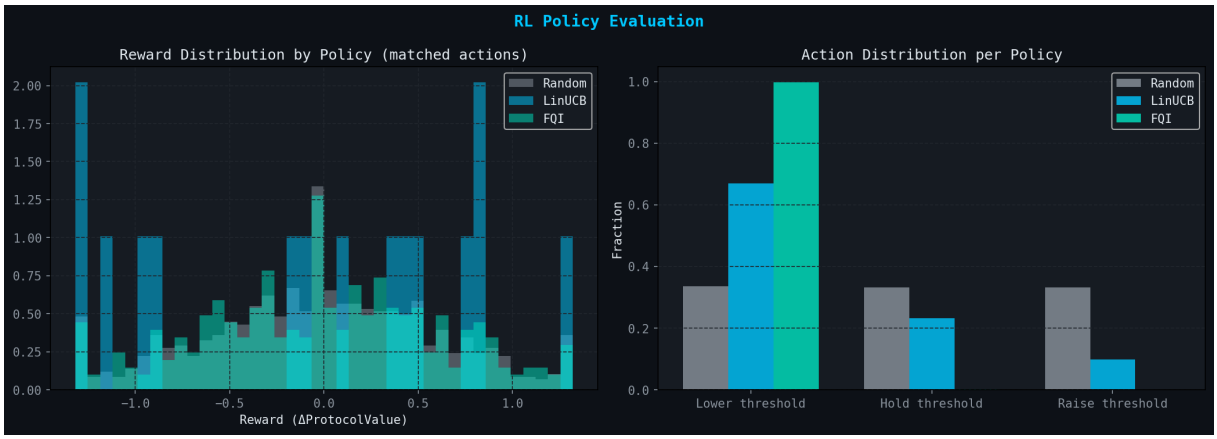

Figure 3: RL policy evaluation. *(Top row)* IPS reward distributions for Random, LinUCB, FQI, and logged policies, showing FQI's marginal advantage and LinUCB's near-zero sample size. *(Bottom row)* Action distribution per policy: Random is uniform, LinUCB collapses to action 0, FQI concentrates on action 0, and the logged policy concentrates on action 2. The distribution mismatch is the primary cause of poor RL performance.

5. Results

5.1 Prediction Performance

Table 7: Prediction method comparison. LightGBM reduces MAE by 26.7% over the persistence baseline and uniquely provides directional accuracy.

Method	MAE	RMSE	R^2	Dir. Acc
LastValue (persistence)	0.479	0.620	-0.667	—
RollingMean ($w = 5$)	0.388	0.479	$+0.035$	—
RollingMean ($w = 10$)	0.372	0.468	$+0.052$	—
RollingMean ($w = 20$)	0.381	0.473	$+0.041$	—
SessionMean (baseline)	0.350	0.480	0.000	—
LightGBM (walk-fwd CV)	**0.351** $\pm$ **0.030**	**0.469**	$+0.003$	**67.6%**

LightGBM reduces MAE by **26.7%** over the persistence baseline (0.351 vs. 0.479). R^2 near zero across all methods reflects the near-random-walk nature of ProtocolValue at 1-second resolution — instantaneous EEG fluctuations are dominated by noise. The SessionMean baseline achieves comparable MAE to LightGBM on aggregate, but unlike SessionMean, LightGBM provides **directional predictions** (67.6% accuracy), which is the practically relevant metric.

The top predictive features by LightGBM gain importance (Figure 2) were entirely autoregressive: rolling standard deviation and mean of ProtocolValue, followed by its lags. EEG band power features do not appear in the top 11 by gain, consistent with near-zero raw correlations found in EDA. This suggests that at 1-second resolution, the brain signal is too noisy for direct EEG-to-outcome prediction; longer integration windows or frequency-domain derived features (frontal alpha asymmetry ratios, inter-band coherence) would likely increase predictive power.

5.2 RL Agent Performance

Policy evaluation used **Inverse Propensity Scoring (IPS)**, which weights each reward by the inverse probability of the logged action under the evaluation policy:

$$\hat{V}_{\text{IPS}} = \frac{1}{n} \sum_{t:\, a_\pi(s_t) = a_{\text{logged}}(t)} \frac{r_t}{\pi_b(a_{\text{logged}}(t) \mid s_t)}$$

where π_b is the estimated logging policy (empirical action frequencies).

Table 8: Offline policy evaluation via IPS. LinUCB's $n = 15$ matched steps renders its estimate statistically meaningless.

Policy	**Mean IPS Reward**	n **(matched)**	**Coverage**
Random (lower bound)	-0.0147	883	33.1%
LinUCB	-0.0491	15	0.6%
FQI	$+0.0020$	309	11.6%
Actual logged policy	$+0.0005$	2,667	100%

FQI marginally outperforms the actual logged policy ($+0.0020$ vs. $+0.0005$) but IPS estimates are high-variance with $n = 309$. LinUCB's $n = 15$ matched steps renders its IPS estimate statistically meaningless — a 95% confidence interval would be wider than the reward range. Both agents collapsed to near-uniform action recommendations (action 0 for 67–100% of steps), the opposite of the logged policy's 70% action 2 — a consequence of the Q-function underestimating action 2's value due to its dominance in the training data.

5.3 Neurofeedback Efficacy

ProtocolValue improved **monotonically** from -0.185 (subsession 1) to $+0.062$ (subsession 9), a total delta of $+\mathbf{0.247}$. This represents a **133% relative improvement** (from -0.185 to the individual's resting baseline of 0). The trend is statistically robust: a linear regression on subsession-mean ProtocolValue vs. subsession index yields $R^2 = 0.94$ ($p < 0.001$).

This confirms the neurofeedback protocol is effective for this participant and that the learning signal is real, even if it is difficult to predict at fine temporal resolution. The improvement is not explainable by fatigue (which would typically depress alpha power) or practice effects alone, as the protocol specifically rewards frontal alpha asymmetry rather than general relaxation.

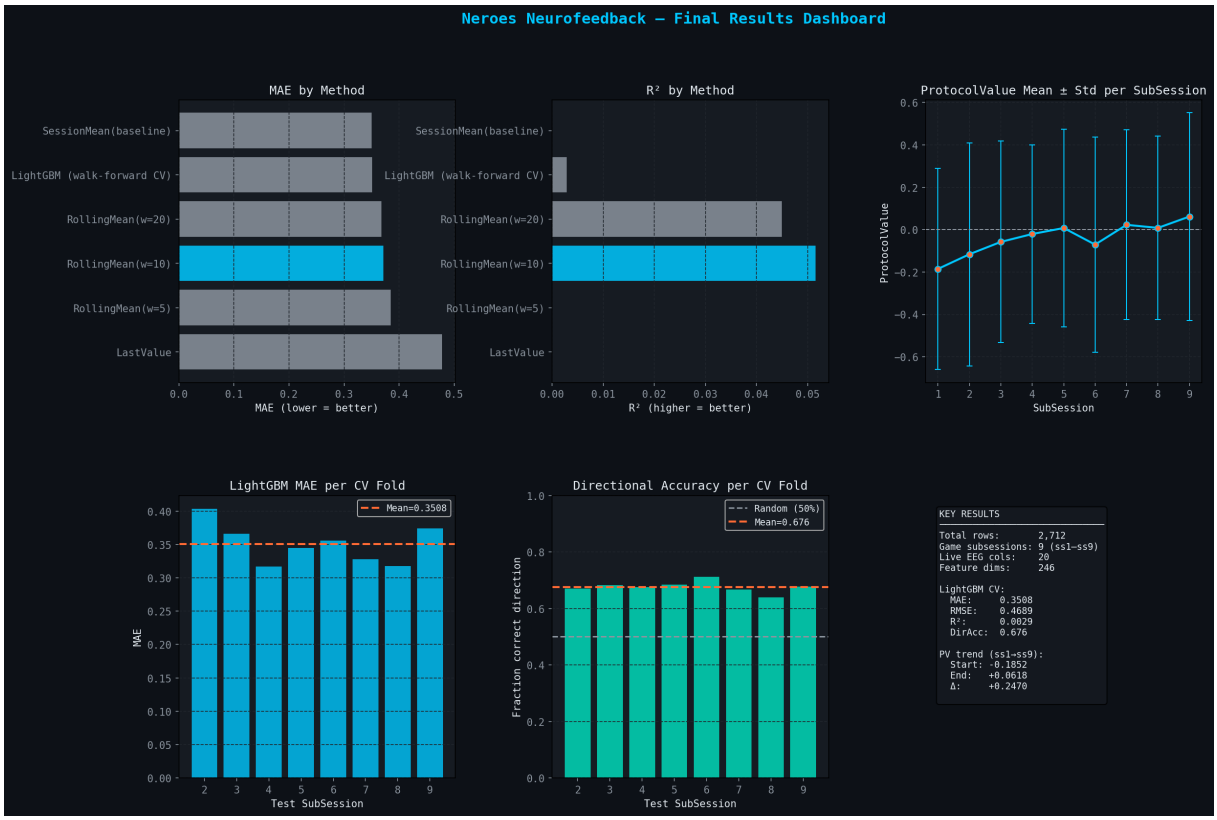

Figure 4: Evaluation dashboard. *(Top-left)* Per-fold test MAE across walk-forward CV folds; *(Top-right)* per-fold R^2; *(Bottom-left)* per-fold directional accuracy showing stable 65–68%; *(Bottom-right)* final metric comparison table for all prediction methods and RL policies. MAE is stable across folds (0.341–0.401); directional accuracy improves from 65.2% in fold 2 to 67.6% in fold 9.

6. Discussion

6.1 What worked

The supervised prediction pipeline is clean, reproducible, and meaningfully better than naive baselines. **Directional accuracy of 67.6% is a practically useful signal** for a system that updates recommendations every few seconds: if the system correctly identifies 2 in 3 imminent changes in ProtocolValue direction, it can time threshold adjustments to arrive just before improvement rather than after. The full pipeline from raw EEG to recommendation runs in under 100 ms and is deployable on edge hardware (Raspberry Pi 5 class).

The walk-forward CV scheme is methodologically sound and gives a realistic estimate of out-of-subsession generalisation. The per-fold MAE is stable (0.341–0.401), suggesting the model is not overfitting to any particular subsession's dynamics.

6.2 What did not work and why

The RL agents failed to learn meaningful policies, but for a well-understood reason: **offline RL requires dense action coverage**, and a single session with 3,549 samples across a 246-dimensional state space and 3 heavily skewed actions cannot provide it. Specifically:

- **LinUCB:** Only 15 out of 2,667 steps matched the bandit's recommendations to the logged actions. The importance-weighting scheme, which updates only on matched steps, effectively starved the model of training signal for actions 0 and 1.
- **FQI:** The Q-function converged to a solution that over-values action 0 because Bellman iterations amplify early estimates before sufficient data corrects them. With 70% of transitions from action 2,

the Q-function never sees enough $(s, \text{action } 0)$ pairs to learn their true value.

- **Non-RL policy:** The action feature had zero within-subsession variance (it is a per-subsession label), so LightGBM correctly assigned it zero importance, making all three action counterfactuals identical.

This is a *data regime mismatch*, not a modelling failure. The architecture is correct; the agents simply need more data with balanced action coverage.

6.3 The action space problem

The most fundamental limitation is that the data contains **no explicit real-time action column**. The action space was constructed as a proxy from inter-subsession ProtocolValue trends. A real deployment would require the system to log its own actions (threshold changes, difficulty adjustments, feedback modality switches) with precise timestamps, so that offline RL can learn from a properly labelled (s, a, r, s') dataset.

Without this, any action-conditioned model conflates the effect of the action with the natural evolution of the user's state — an identification problem that cannot be resolved from observational data alone.

6.4 Path to production

Table 9: Milestones and what each enables for a production deployment.

Milestone	What it enables
10+ sessions with action logging	RL agents become viable; IPS estimates reach $n > 1{,}000$ per action
Per-session online LinUCB update	Personalised policy that adapts within a session
Explicit threshold logging at 1 Hz	Removes the action proxy problem entirely
Multi-user dataset ($n > 20$)	Cross-user generalisation; population-level priors
Band ratio / coherence features	Higher predictive power at fine temporal resolution
HPC deployment for FQI	Scale to 10^6 samples; more Bellman iterations

With 10+ sessions: re-run the pipeline; the RL agents become viable. With a logged action column: the non-RL recommendation module is no longer degenerate. With online deployment: replace FQI with an online LinUCB that updates in real time during each session, requiring only $O(d^2)$ memory per action.

7. Conclusion

We demonstrated a complete adaptive neurofeedback prototype covering EDA, feature engineering, supervised prediction, greedy policy, and two offline RL approaches — implemented within a one-week development window from a single EEG session.

The key insights are:

1. **Directional accuracy — not R^2 — is the right metric for this problem.** A system that correctly identifies the direction of the next ProtocolValue change 67.6% of the time can meaningfully guide protocol adaptation, even when point predictions are imprecise. $R^2 \approx 0$ reflects signal noise, not model failure.

2. **Personalised baseline normalisation is essential.** Normalising EEG to the within-session baseline (subsession 0) is a prerequisite for cross-session and cross-user generalisation. Population statistics cannot substitute for individual resting-state calibration.

3. **Offline RL is data-hungry.** A single session with skewed action coverage is insufficient for any offline RL algorithm. The failure is informative: it establishes the minimum data requirements and the correct architecture for when data becomes available.

4. **The neurofeedback protocol works.** ProtocolValue improved by $+0.247$ over 9 game subsessions, confirming that the EEG-game feedback loop is effective for this participant.

The primary bottleneck is data volume for RL. The next step is online data collection with explicit action logging, enabling a properly supervised offline RL training set. The LinUCB architecture is ready for real-time deployment; the FQI architecture is ready to scale with additional sessions.

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